

Seeds for Life: Modeling the Cometary Dynamics of Epsilon Eridani with N-Body Simulations

Riley Hebert, Virginie Faramaz-Gorka
Steward Observatory, The University of Arizona, Tucson, AZ (USA)

rileyhebert@arizona.edu



Motivation

Mapping the dynamics of exoplanetary systems allows us to gain insights on one possible delivery method of the ingredients for life: comets originating beyond the ice line on eccentric orbits traveling into the Habitable Zone. Systems of interest must have dust near the star and an asteroid belt analogue as well as a Kuiper belt analogue. One such system is **Epsilon Eridani (Eps Eri)**.
The goal of this project was to determine if the evaporating comets sustaining the exozodi formed with the chemical compounds necessary for living organisms to arise.

Background

Eps Eri is located ~10.5 ly from Earth and consists of a K2 type star orbited by a gas giant, **Eps Eri b**, and at least one other planet farther out, **Eps Eri c**.

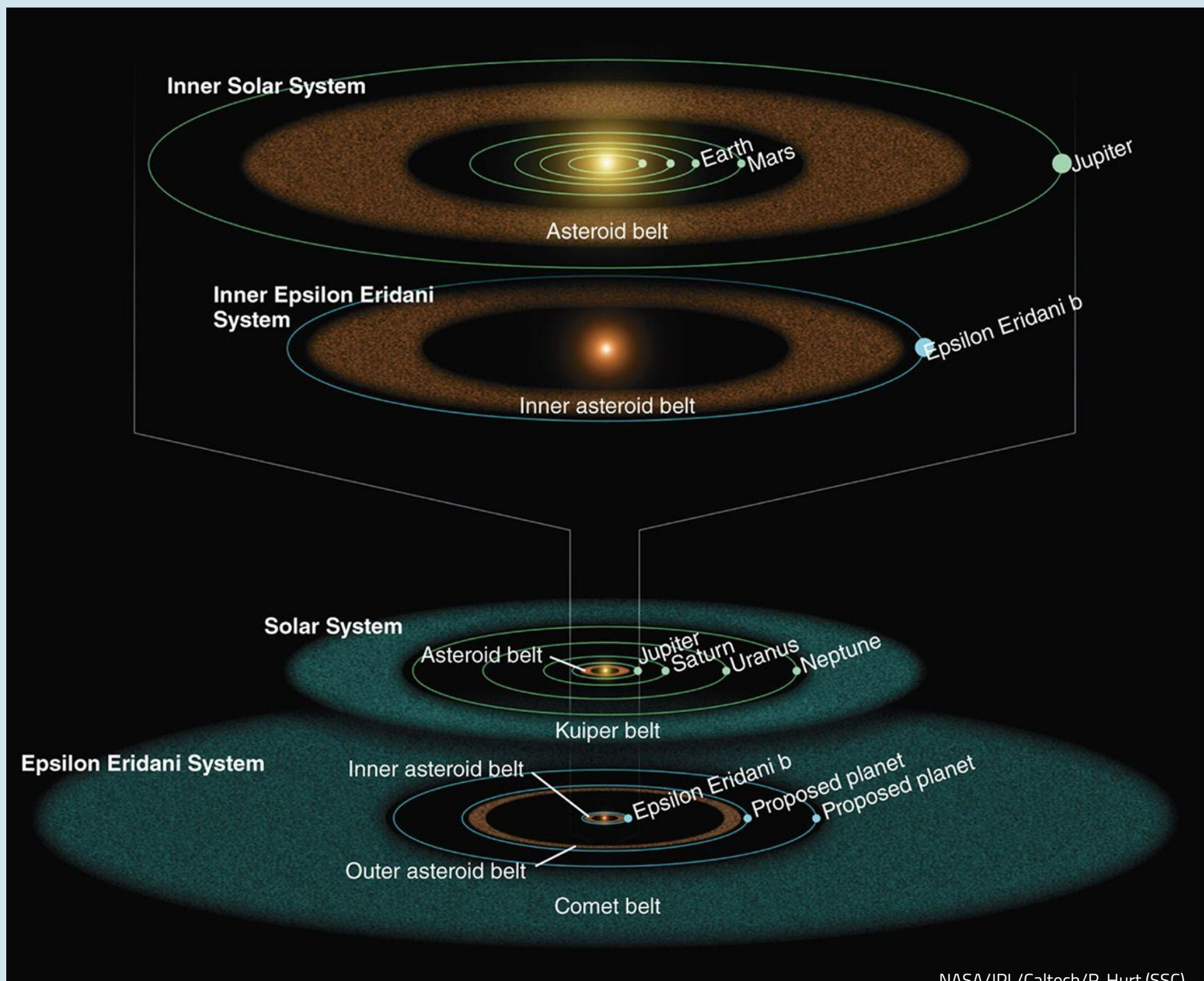


Fig. 1: Eps Eri compared to the Solar System

The system has a warm reservoir of comets within Eps Eri b's orbit as well as a cold reservoir beyond the **ice line**. In addition, this system has a large amount of **exozodiacal dust (exozodi)** that cannot be created in situ and must be sustained by a mechanism delivering dust that formed elsewhere to the inner system. It is most likely being replenished by the evaporation of comets passing near the star. The dynamics of the comets in the warm and cold reservoirs were modeled with an **N-body Simulation** in order to determine where these comets originate. In particular, comets that formed in the cold belt may carry **volatiles** into the inner system, which are required for life to arise. Thus, **Eps Eri was modeled with an N-body simulation that spotted particles within 1AU of the star and the results were analyzed with Python code in order to determine if it has the necessary components for life.**

Analyzing the N-Body Simulation

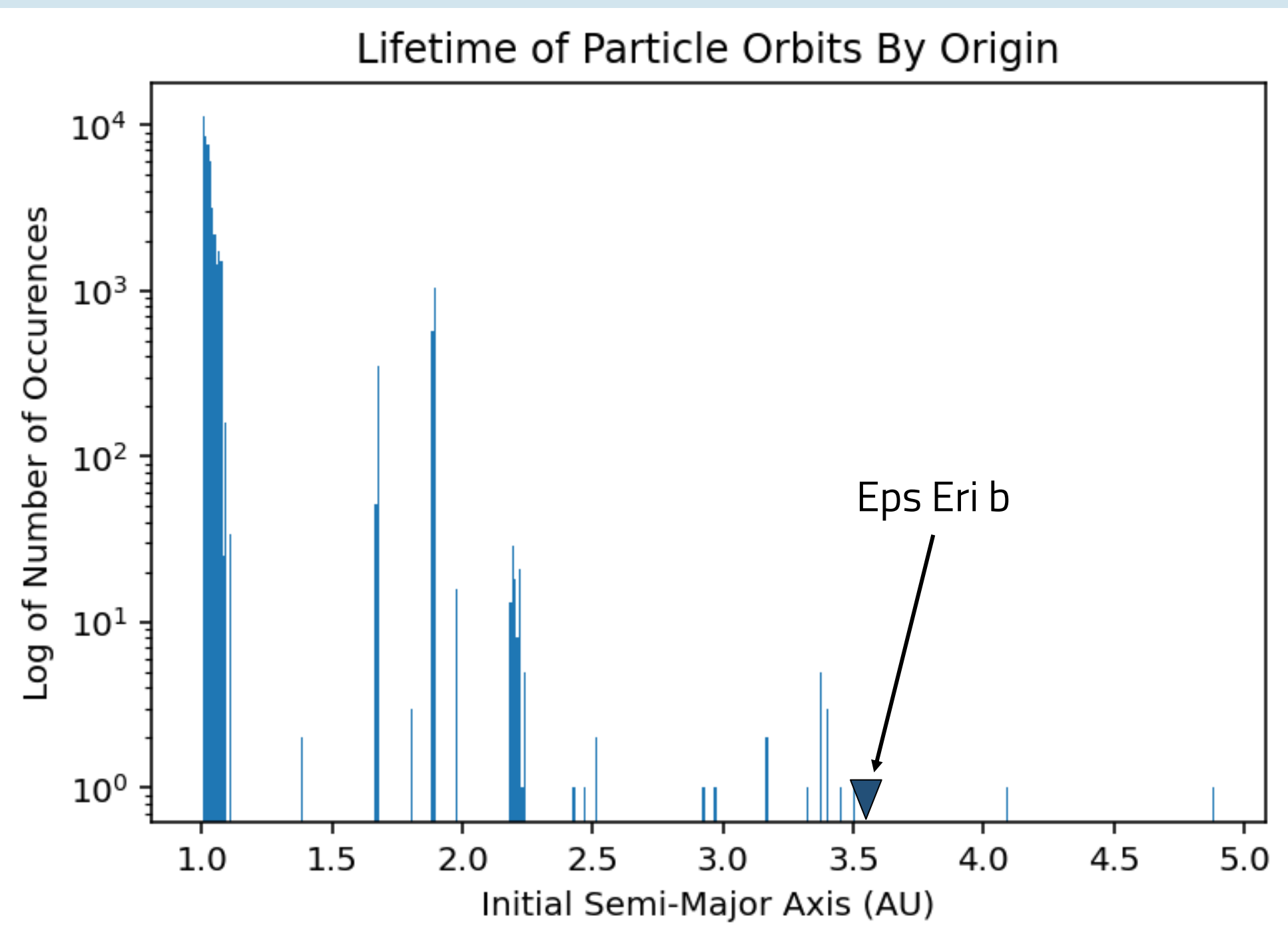


Fig. 2

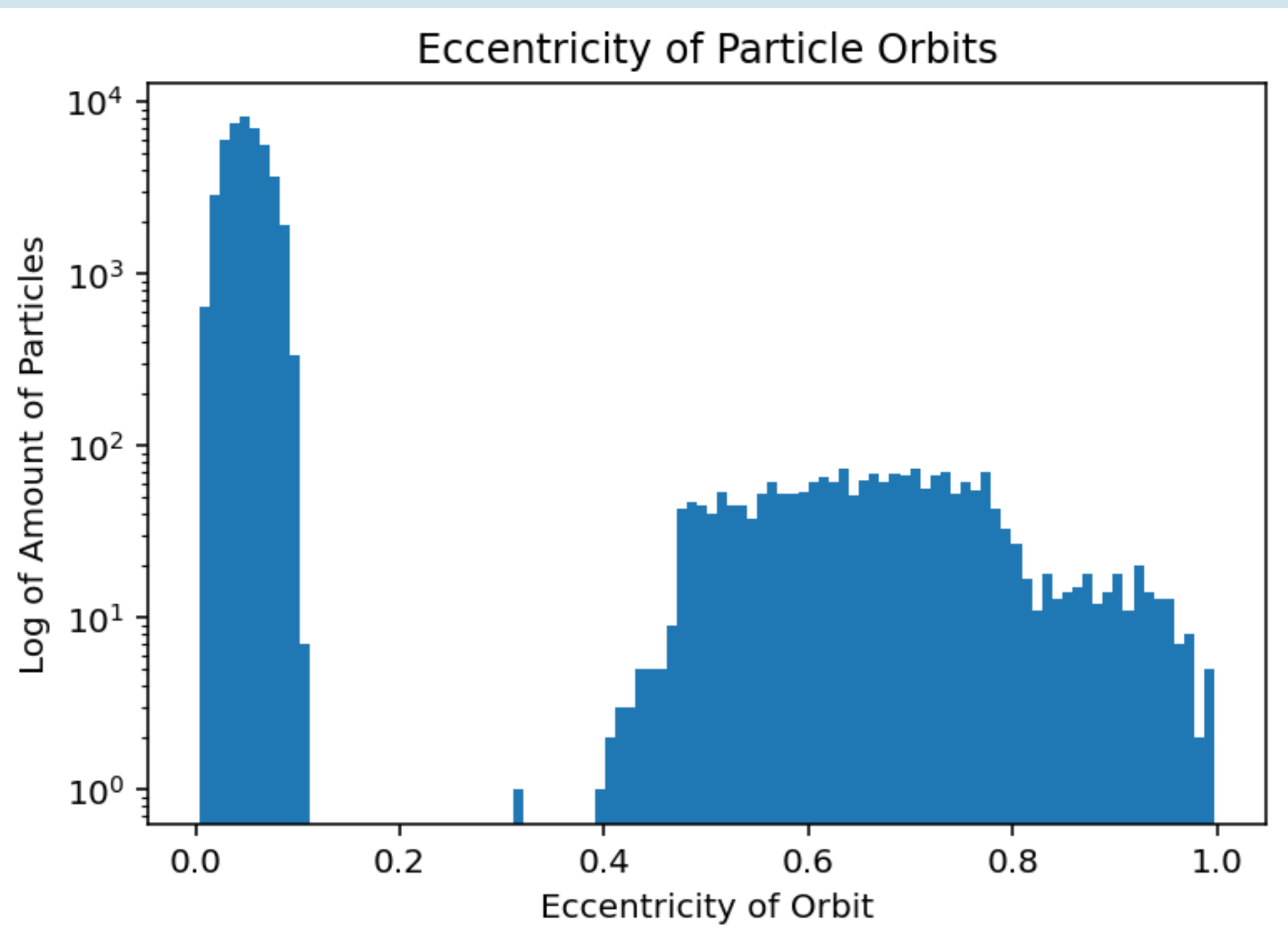


Fig. 3

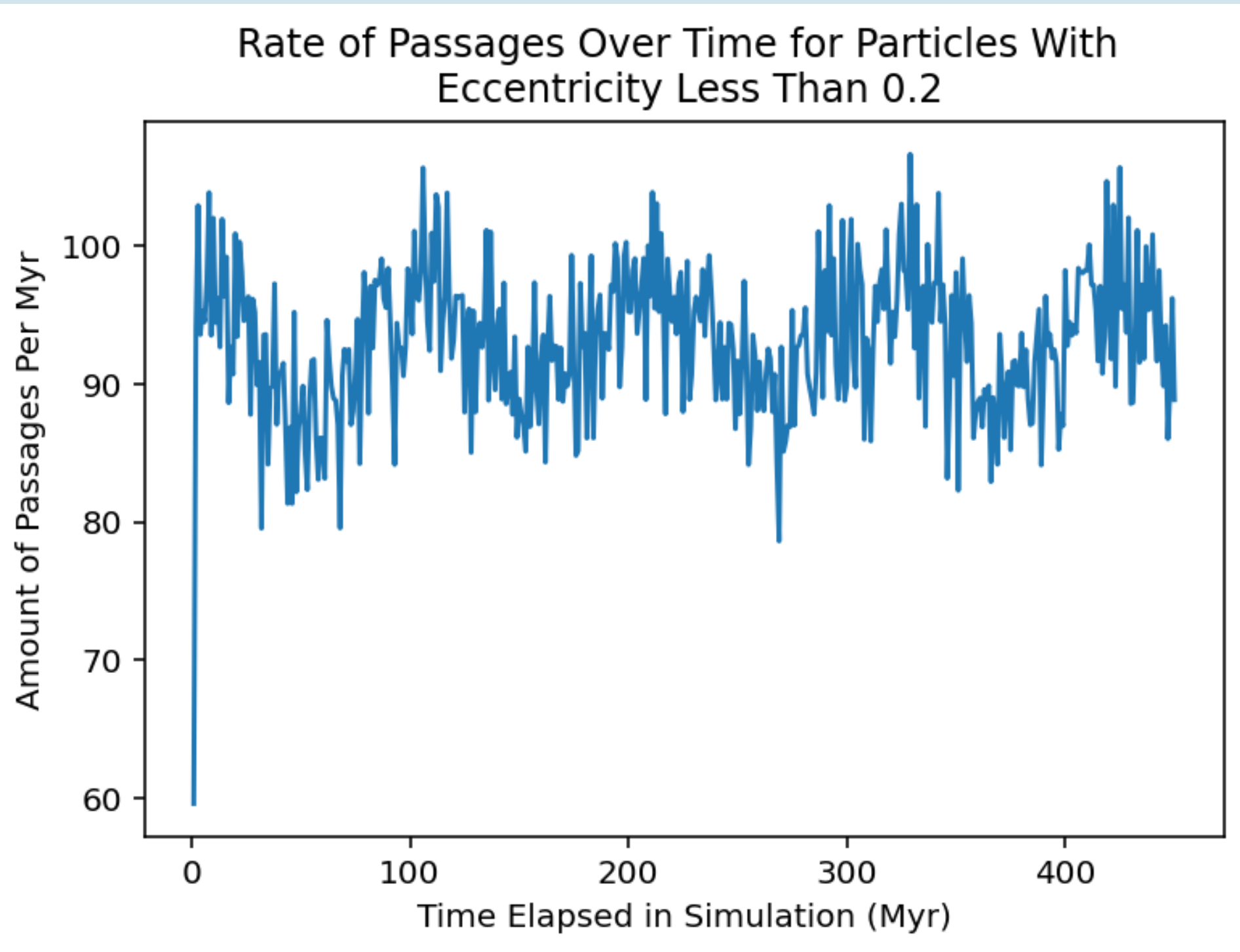


Fig. 4

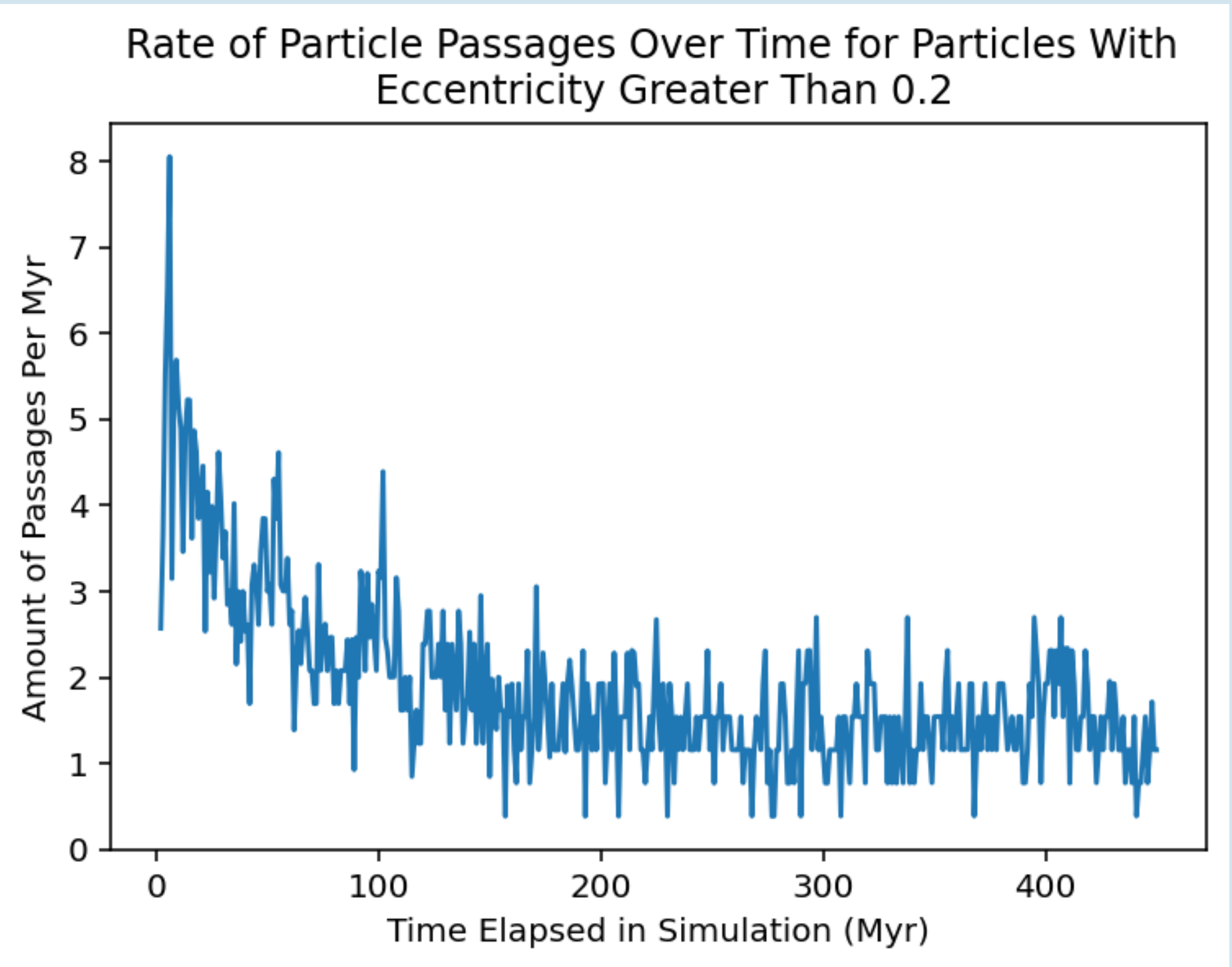


Fig. 5

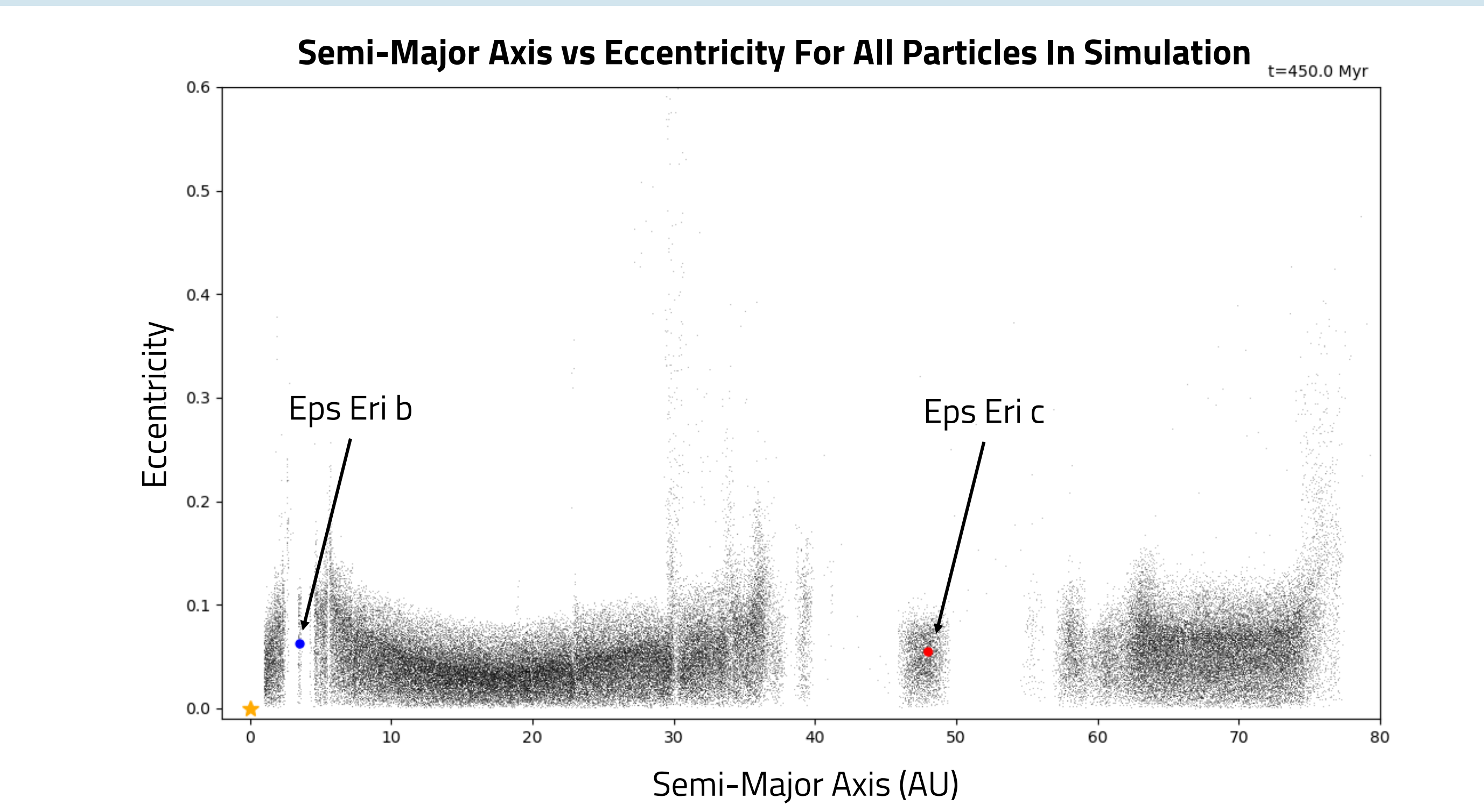


Fig. 6

Results

- Figure 2 shows that the majority of comets that are spotted originate within Eps Eri b's orbit..
- Figure 3 shows that most of the particles spotted fall into two distributions, one of eccentricity less than 0.2 and a smaller population with eccentricity greater than 0.2.
- Figure 4 graphs the rate of passages per year corresponding to particles in the population with eccentricity less than 0.2. The sinusoidal shape of the curve indicates that the orbits oscillate around the star but are consistent overall.
- Figure 5 in turn graphs the rate of passages per year corresponding to the population with eccentricity greater than 0.2. The decreasing exponential curve implies that over time some mechanism prevents higher-eccentricity particles from passing by the star.
- Figure 6 shows the eccentricity as a function of the semi-major axis for all particles in the simulation. It demonstrates that very few of the comets available pass through the inner system. From Figure 2, those that do almost exclusively originate in the small population to the left of Eps Eri b.

Conclusions

Therefore, the comets that replenish the exozodi are from the warm reservoir, within Eps Eri b's orbit. In other words, **Eps Eri b's gravitational influence acts as a shield, preventing comets originating from the cold belt from reaching the inner system.**

Because no ices are transported, **it is unlikely that Eps Eri's Habitable Zone has received the necessary components for life,** at least through this mechanism. However, this experiment allows us to better understand the shielding behaviors large planets can demonstrate, and early models of the Solar System must account for that behavior.

Acknowledgements

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